

Change of measure and Girsanov's Theorem

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1 Translational symmetry of a trajectory

A trajectory is a property of a physical object.

Definition A physical object with a quantity X

Definition Representation of X in the reference frame (P)
 $dX = \mu dt + \sigma dW^P$.

Definition Representation of X for a drifting (with μ) observer (Q)
 $d'X = \sigma dW^Q$.

Observation Mean

$$\begin{aligned}\mathbb{E}^P[X(T) - X(t)] &= \mathbb{E}^P\left[\int_t^T dX\right] = \mu(T - t) \\ \mathbb{E}^Q[X(T) - X(t)] &= \mathbb{E}^Q\left[\int_t^T d'X\right] = 0.\end{aligned}$$

Observation Second Moment

$$\begin{aligned}\mathbb{E}^P[(X(T) - X(t))^2] &= \mathbb{E}^P\left[\left(\int_t^T dX\right)^2\right] = \sigma^2(T - t) + \mu^2(T - t)^2 \\ \mathbb{E}^Q[(X(T) - X(t))^2] &= \mathbb{E}^Q\left[\left(\int_t^T d'X\right)^2\right] = \sigma^2(T - t)\end{aligned}$$

Abstraction Common physical intuition on Brownian motion

Note :

- Physical quantity is something conserved under transformation
- Martingale under Q is $\mathbb{E}^Q[X(T)|\mathcal{F}_t] = X(t)$, which is conserved amount over time transform.
- $\mathbb{E}^P[X(T)|\mathcal{F}_t] \neq X(t)$, so P is less useful frame.

2 Change of measure for probability distribution

Is probability distribution a physical object?

Observation $\frac{dP}{d(X(T)-X(t))} = P^P(X(T) - X(t)) = N(\mu(T - t), \sigma^2(T - t))$

Observation $\frac{dQ}{d(X(T)-X(t))} = P^Q(X(T) - X(t)) = N(0, \sigma^2(T-t))$

Abstraction Probability space $(\Omega, \mathcal{F}, P)$, $(\Omega, \mathcal{F}, Q)$

Observation $\frac{dP}{dQ} = \frac{P^P(X(T)-X(t))}{P^Q(X(T)-X(t))}$

$$= \frac{\exp \frac{-(X(T)-X(t)-\mu(T-t))^2}{2\sigma^2(T-t)}}{\exp \frac{-(X(T)-X(t))^2}{2\sigma^2(T-t)}} = \exp \frac{-(X(T)-X(t)-\mu(T-t))^2 + (X(T)-X(t))^2}{2\sigma^2(T-t)}$$

3 GBM case

Definition $X = \log S$

P: reference frame

Q: drifting with $\mu S dt$. $dS = \mu S dt + \sigma S dW^P$.

Definition Representation of S in the reference frame (P)

$$dS = S(\mu dt + \sigma dW^P).$$

Definition Representation of S for a drifting(with μ) observer (Q)

$$d'S = S(\sigma dW^Q)$$

Observation

Under $\mathcal{F}_t$,

$$\frac{dP}{d(X(T)-X(t))} = p^P(X(T) - X(t)) = N((\mu - \frac{1}{2}\sigma^2)(T-t), \sigma^2(T-t))$$

$$\frac{dQ}{d(X(T)-X(t))} = p^Q(X(T) - X(t)) = N((-\frac{1}{2}\sigma^2)(T-t), \sigma^2(T-t))$$

Observation

$$\begin{aligned} \frac{dP}{dQ} &= \frac{P^P(X(T)-X(t))}{P^Q(X(T)-X(t))} = \frac{\exp \frac{-(X(T)-X(t)-(\mu-\frac{1}{2}\sigma^2)(T-t))^2}{2\sigma^2(T-t)}}{\exp \frac{-(X(T)-X(t)-(-\frac{1}{2}\sigma^2)(T-t))^2}{2\sigma^2(T-t)}} \\ &= \exp \frac{-(X(T)-X(t)-(\mu-\frac{1}{2}\sigma^2)(T-t))^2 + (X(T)-X(t)-(-\frac{1}{2}\sigma^2)(T-t))^2}{2\sigma^2(T-t)} \\ &= \exp \frac{-(\Delta X - (\mu - \frac{1}{2}\sigma^2)\Delta t)^2 + (\Delta X - (-\frac{1}{2}\sigma^2)\Delta t)^2}{2\sigma^2\Delta t} \\ &= \exp \frac{2\Delta t\Delta X(\mu - \frac{1}{2}\sigma^2) + \sigma^2\Delta X\Delta t - \mu^2\Delta t^2 + \sigma^2\mu\Delta t^2}{2\sigma^2\Delta t} \\ &= \exp \frac{2\mu\Delta X\Delta t - (\mu^2 - \mu\sigma^2)\Delta t^2}{2\sigma^2\Delta t} = \exp \left(\frac{\mu}{\sigma^2}\Delta X - \frac{\mu^2 - \mu\sigma^2}{2\sigma^2}\Delta t \right) \\ &= \exp \left(\frac{\mu}{\sigma^2}((\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\Delta W^P) - \frac{\mu^2 - \mu\sigma^2}{2\sigma^2}\Delta t \right) \\ &= \exp \left((\frac{\mu^2}{2\sigma^2})(\Delta t) + \frac{\mu}{\sigma}(\Delta W^P) \right) \end{aligned}$$

4 Abstraction : Girsanov theorem

- Girsanov thm

$$Z_t = \frac{dQ}{dP} \Big|_{\mathcal{F}_t} = \exp \left(- \int_0^t \theta_s dW_s^P - \frac{1}{2} \int_0^t \theta_s^2 ds \right), \theta_s = \frac{\mu}{\sigma}$$

- Consistency check

$$\frac{dP}{dQ} \frac{dQ}{dP} = \exp \left(+(\frac{\mu^2}{2\sigma^2})(\Delta t) + \frac{\mu}{\sigma}(\Delta W^P) - \int_0^t \theta_s dW_s^P - \frac{1}{2} \int_0^t \theta_s^2 ds \right) = 1$$